FraudShield AI | Executive Summary

Capstone project • Educational prototype • Synthetic data only

1. Problem and value proposition

Currently, transaction review teams struggle to prioritize potentially suspicious activity at scale. FraudShield AI uses supervised machine learning to assign a review-priority score from transaction attributes. Its intended outcome is to help reviewers focus attention on higher-risk records, not to automate punitive decisions.

2. Data and preparation

The demonstration dataset contains 6,000 synthetic transactions with six numerical/binary input features: transaction amount, hour of day, transactions in the past hour, international flag, new-device flag, and account age in days. Labels are simulated, not confirmed fraud cases. Data were checked for missing values and class distribution. A stratified 75/25 train/test split preserves label proportions.

3. Model and evaluation

A Random Forest classifier with class weighting was trained using a fixed random seed. A 0.35 review threshold was used to illustrate the trade-off between missed suspicious transactions and unnecessary investigations. The held-out synthetic test results are shown below. These results are not estimates of performance on real banking data.

Metric	Synthetic test result
Precision	9.9%
Recall	53.6%
F1	16.7%
Average precision	11.1%

4. Product experience and deployment

A Streamlit dashboard accepts a transaction profile and displays the model risk score and whether the transaction crosses the review threshold. It also shows evaluation metrics and an exploratory view of synthetic data. Run the dashboard locally using the included requirements and README. No public deployment URL is claimed.

5. Impact and evidence

The working prototype demonstrates the full lifecycle: problem framing, data preparation, model training, held-out evaluation, interactive interface, and stakeholder communication. The observed metrics describe only the synthetic test set. Operational benefits such as reduced review time, improved fraud capture, and financial savings remain hypotheses requiring a real-world pilot.

6. Risk management and future scope

Key risks include false positives, missed fraud, biased outcomes, drift, privacy exposure, and overconfidence in simulated metrics. Before real deployment: obtain permissioned representative data, perform temporal validation and subgroup audits, calibrate scores, choose thresholds with business stakeholders, monitor performance, and keep human review and appeals in place.

7. Submission map

GitHub-ready repository: README.md, app.py, train.py, requirements.txt, synthetic dataset, trained model, metrics, and pitch deck. The companion five-slide pitch follows the course guide: hook, data, solution, impact, future scope.

Evaluation rubric alignment

Model performance 30% • Data quality 20% • UI/UX and deployment 20% • Presentation 20% • Problem definition 10%.